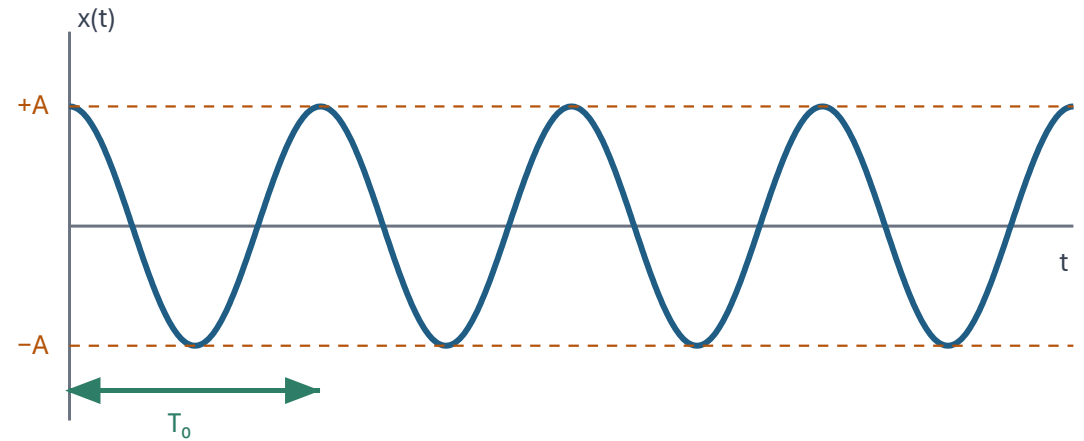


Topic 2

Sinusoids

From cosine waves to complex phasors



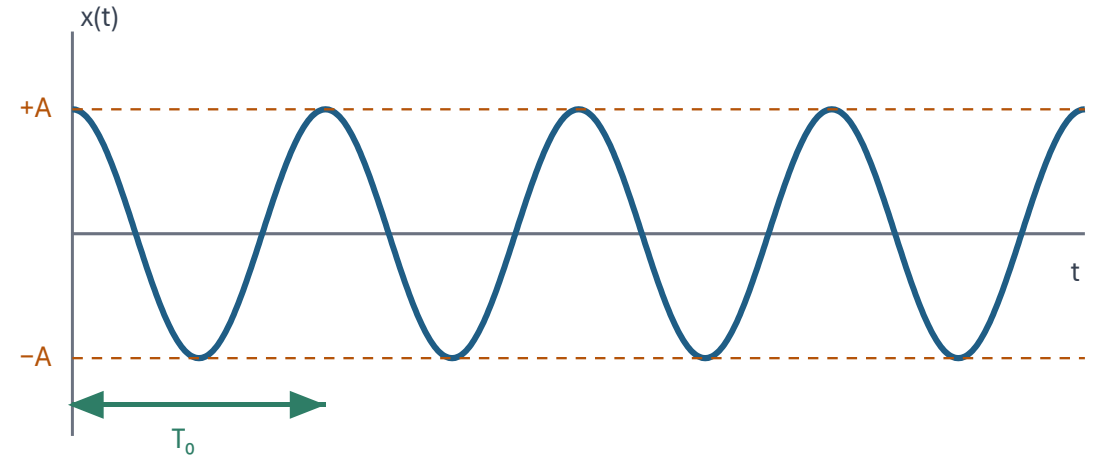
Key idea. Sinusoids are the basic oscillating building blocks of signals. Complex exponentials turn trigonometric manipulations into vector algebra.

What is a sinusoid?

A cosine with a linearly increasing phase

$$x(t) = A \cos(\omega_0 t + \phi)$$

- A : amplitude / peak value
- ω_0 : radian frequency in rad/s
- ϕ : initial phase in radians
- The signal repeats with period T_0

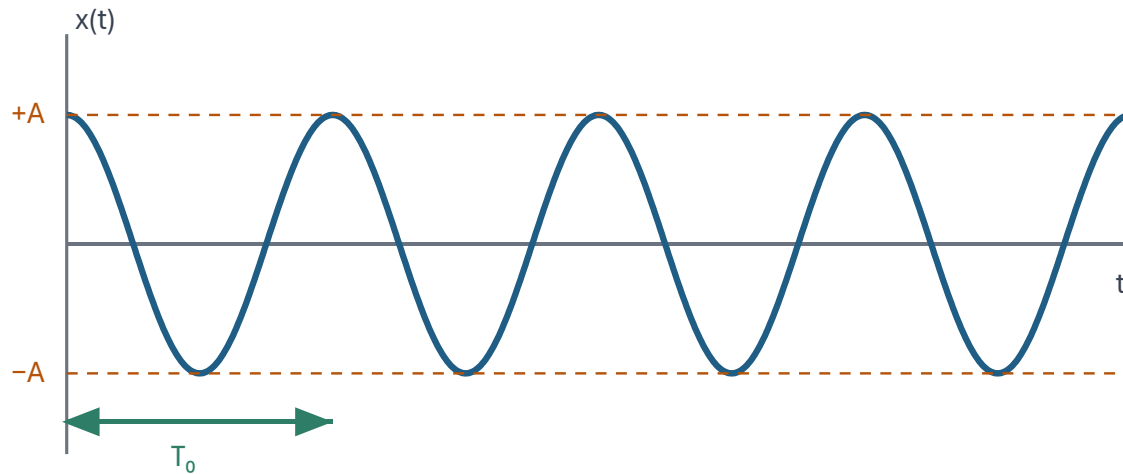


$$f_0 = \frac{\omega_0}{2\pi}, \quad T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$$

A changes vertical scale; ω_0 changes how fast it oscillates; ϕ shifts the waveform horizontally.

Physical motivation: the A-440 tuning fork

A near-pure tone is well modelled by a sinusoid



one cycle every 2.27 ms

$$x(t) = 10\cos(2\pi \cdot 440 \cdot t - 0.4\pi)$$

- Frequency: 440 Hz — A above middle C.
- Period: $1/440$ s $\approx$ 2.27 ms.
- Amplitude controls loudness-related scale, not pitch.
- Initial phase sets where the cycle starts at $t=0$.

Sine and cosine toolbox

Identities used throughout sinusoidal analysis

Phase relation	$\sin(\theta) = \cos(\theta - \pi/2), \cos(\theta) = \sin(\theta + \pi/2)$
Symmetry	$\cos(-\theta) = \cos(\theta), \sin(-\theta) = -\sin(\theta)$
Pythagorean identity	$\sin^2(\theta) + \cos^2(\theta) = 1$
Angle sum	$\cos(\alpha \pm \beta) = \cos\alpha \cos\beta \mp \sin\alpha \sin\beta$
Derivatives	$d \sin\theta/d\theta = \cos\theta, d \cos\theta/d\theta = -\sin\theta$

Use these identities to move between sine/cosine forms and to derive phasor relationships.

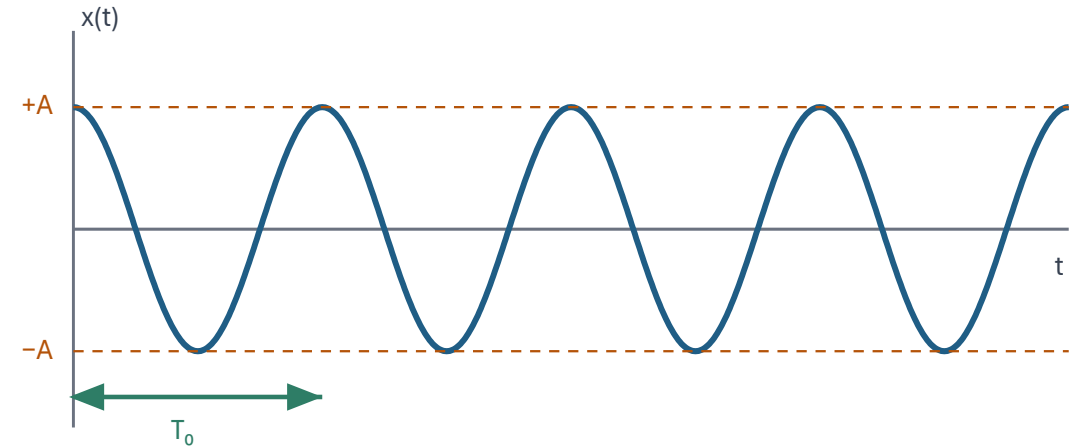
Frequency and period

Two equivalent ways to describe how fast a sinusoid oscillates

$$x(t + T_0) = x(t)$$

$$\omega_0 T_0 = 2\pi \quad \Rightarrow \quad T_0 = \frac{2\pi}{\omega_0}$$

$$f_0 = \frac{1}{T_0}$$



Example

A 440 Hz sinusoid completes 440 cycles each second. The period is approximately 2.27 ms.

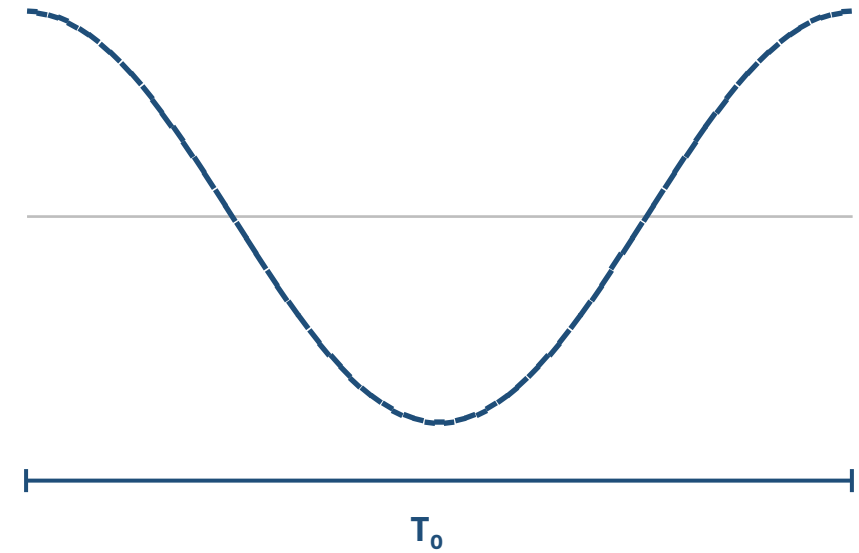
Example 1: frequency and period

Read sinusoidal parameters from the formula

$$x(t) = 2 \cos(0.4\pi t)$$

Find A , ω_0 , f_0 , and T_0 .

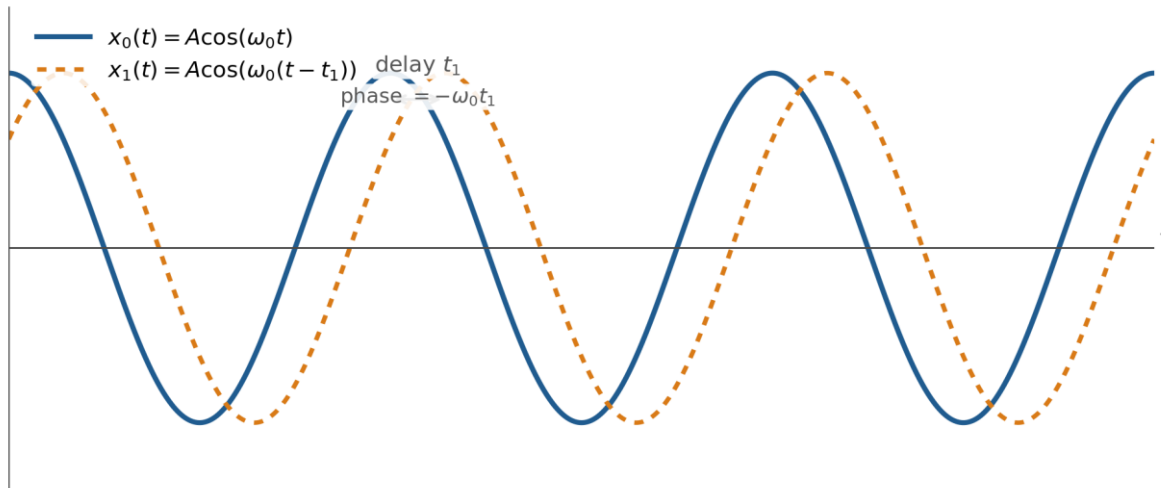
- 1 Amplitude is the coefficient of the cosine: $A = 2$.
- 2 Compare with $A \cos(\omega_0 t + \phi)$: $\omega_0 = 0.4\pi$ rad/s.
- 3 Convert to cyclic frequency: $f_0 = \omega_0/(2\pi) = 0.2$ Hz.
- 4 Period: $T_0 = 1/f_0 = 5$ s.



one cycle = 5 s

Phase shift and time shift

A time delay appears as a negative phase shift



$$A \cos(\omega_0(t - t_1)) = A \cos(\omega_0 t - \omega_0 t_1)$$

- If $t_1 > 0$: the waveform moves right; it is delayed.
- The corresponding phase is $\phi = -\omega_0 t_1$.
- Phase is ambiguous modulo 2π .
- A common convention is $-\pi < \phi \leq \pi$.

$$t_1 = -\frac{\phi}{\omega_0}$$

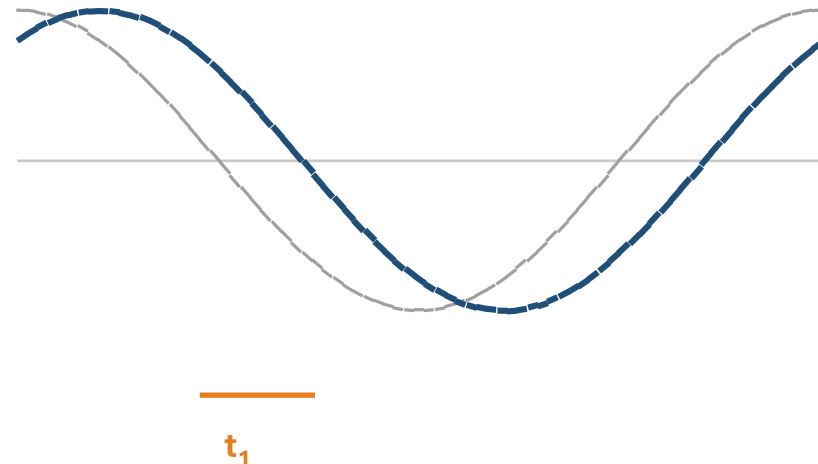
Example 2: phase and delay

A negative phase corresponds to a positive delay

$$x(t) = 3 \cos(20\pi t - 0.2\pi)$$

Rewrite it as $A \cos[\omega_0(t - t_1)]$.

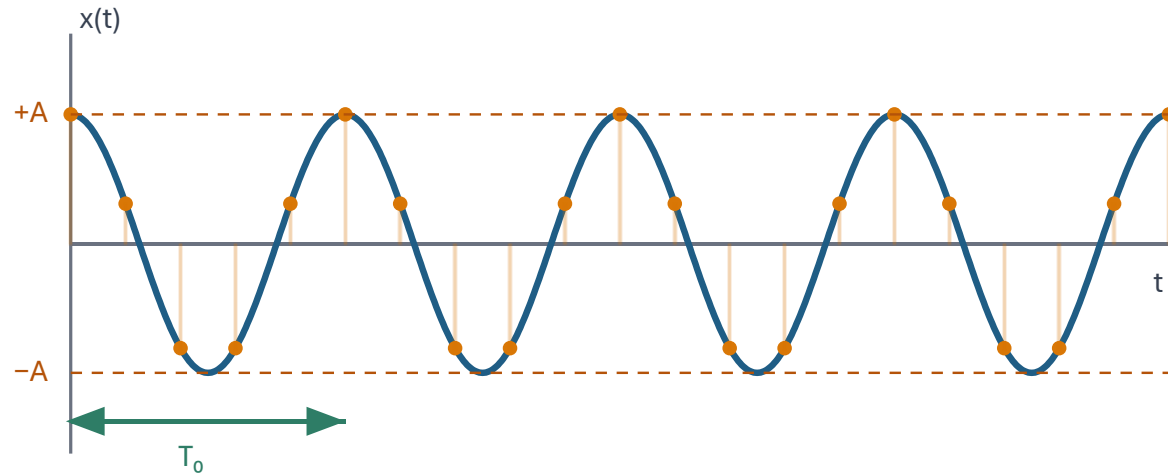
- 1 Identify $\omega_0 = 20\pi$ rad/s and $\phi = -0.2\pi$.
- 2 Use $\phi = -\omega_0 t_1$, so $t_1 = -\phi/\omega_0$.
- 3 $t_1 = 0.2\pi / 20\pi = 0.01$ s = 10 ms.
- 4 Therefore $x(t) = 3 \cos[20\pi(t - 0.01)]$.



delay: $t_1 = 10$ ms

Sampling and plotting sinusoids

A continuous-time sinusoid becomes a sequence evaluated at $t_n = nT_s$



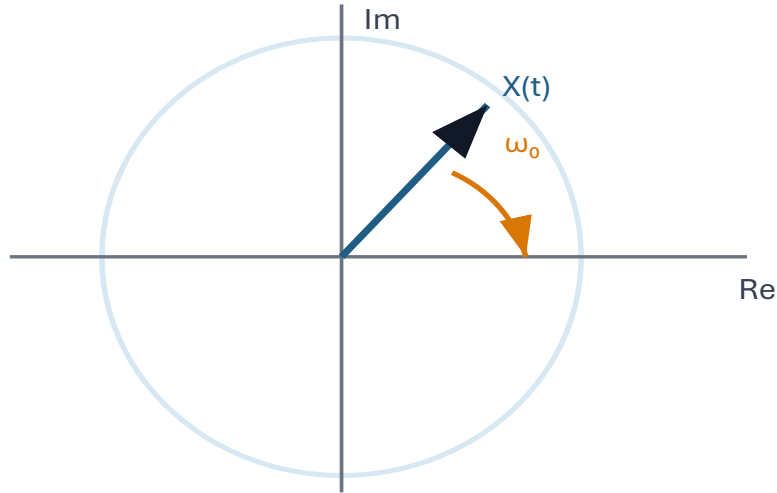
Smaller $T_s \Rightarrow$ more samples per cycle $\Rightarrow$ smoother plot and better time resolution.

$$x[n] = x(nT_s) = A \cos(\omega_0 n T_s + \phi)$$

```
n = np.arange(-7, 6)
Ts = 0.005
tn = n * Ts
xn = 20*np.cos(80*np.pi*tn - 0.4*np.pi)
plt.plot(tn, xn)
```

Complex numbers as vectors

The polar form makes rotations and scaling transparent



$$z = x + jy = re^{j\theta}$$

- Magnitude r is the vector length.
- Angle θ is measured from the real axis.
- Multiplication multiplies magnitudes and adds angles.
- This is why complex numbers are useful for sinusoids.

$$r = |z| = \sqrt{x^2 + y^2}, \quad \theta = \text{atan2}(y, x)$$

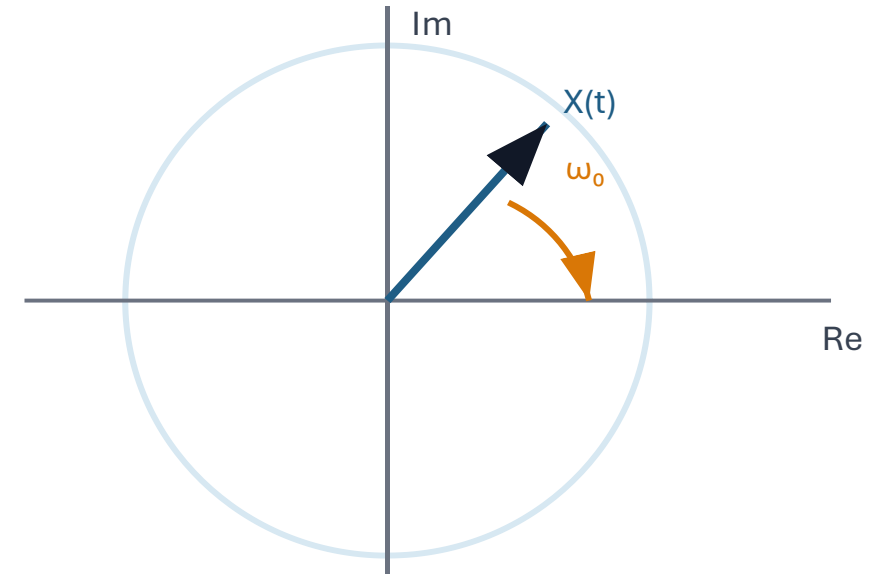
Euler's formula

The bridge between trigonometry and complex exponentials

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$



Interpretation: $e^{j\theta}$ is a unit-length vector at angle θ on the complex plane.

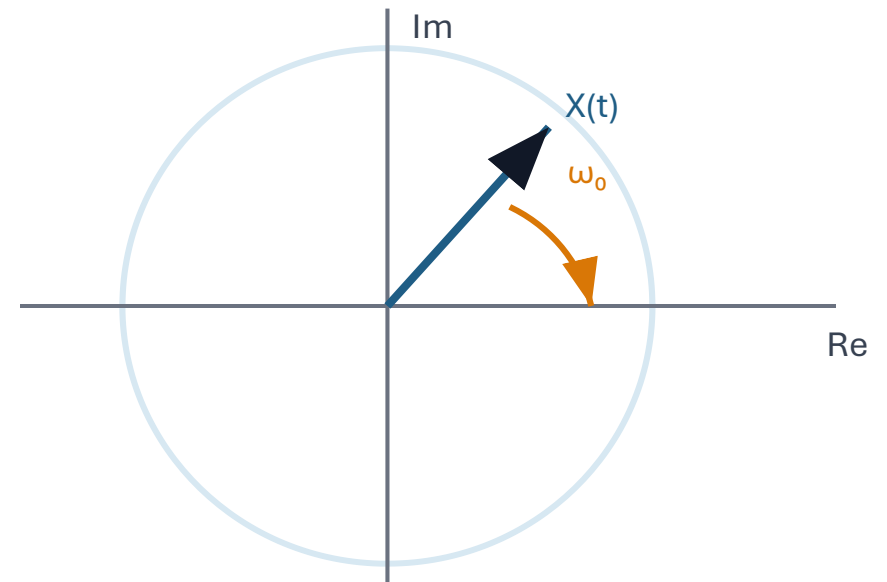
Complex exponential signals

A rotating complex number

$$x_c(t) = X e^{j\omega_0 t}$$

$$X = A e^{j\phi}$$

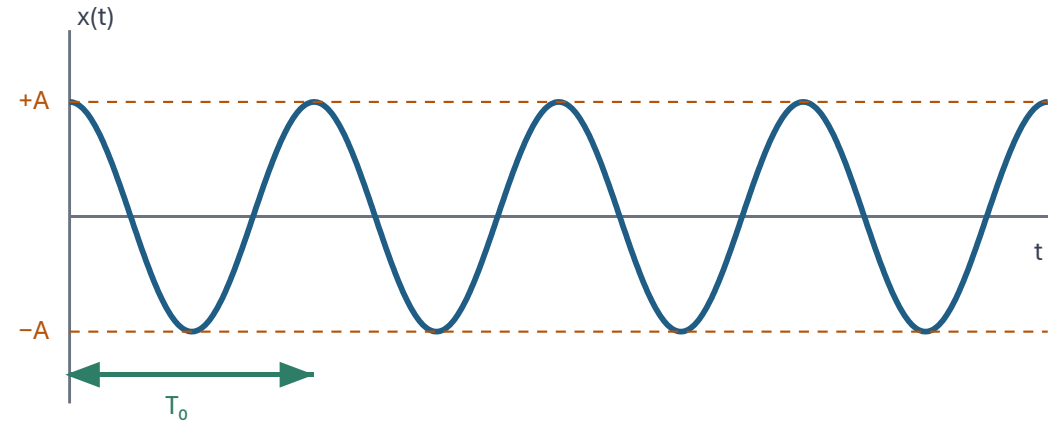
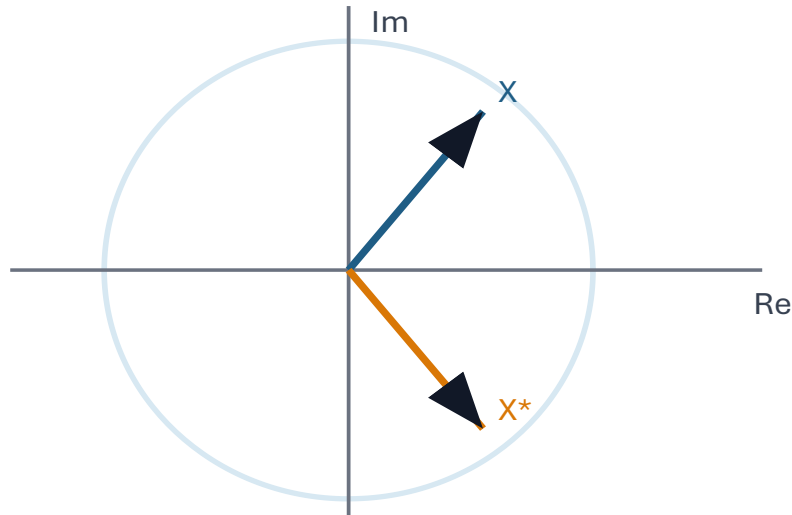
- X is the complex amplitude or phasor.
- $e^{j\omega_0 t}$ rotates the phasor with angular speed ω_0 .
- Positive ω_0 rotates counter-clockwise.
- One full revolution corresponds to 2π radians.



A real cosine as two rotating phasors

Positive and negative frequencies appear as conjugate partners

$$A\cos(\omega_0 t + \phi) = \frac{1}{2}Ae^{j\phi}e^{j\omega_0 t} + \frac{1}{2}Ae^{-j\phi}e^{-j\omega_0 t}$$



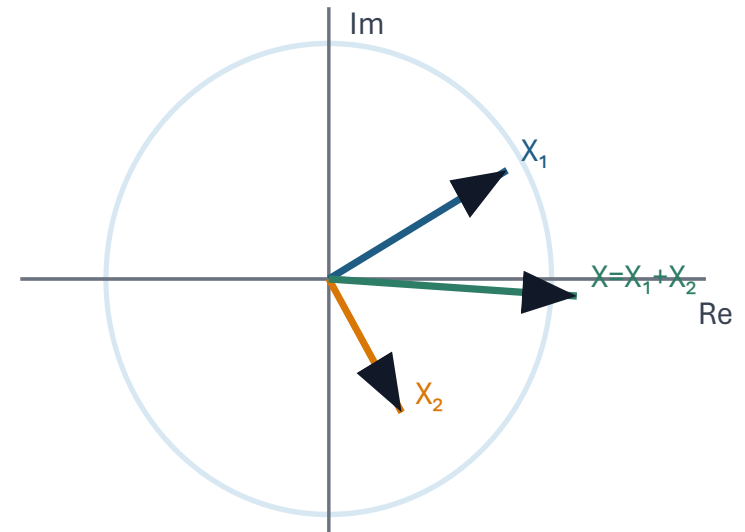
- The conjugate pair makes the sum real.
- One phasor rotates at $+\omega_0$; the other at $-\omega_0$.
- The real part is the observed cosine.

Phasor addition

Same-frequency sinusoids can be combined by adding complex numbers

$$\sum_k A_k \cos(\omega_0 t + \phi_k) = A \cos(\omega_0 t + \phi)$$

- Convert each sinusoid to a phasor $A_k e^{j\phi_k}$.
- Add the phasors as complex numbers.
- Convert the resulting phasor $A e^{j\phi}$ back to cosine form.



$$A e^{j\phi} = \sum_k A_k e^{j\phi_k}$$

Phasor addition: why the rule works

The real-part operator is linear, and all sinusoids share the same rotating factor

1 **Represent each cosine** $x_k(t) = \text{Re}\{X_k e^{j\omega_0 t}\}$

$$X_k = A_k e^{j\varphi_k}$$

2 **Add the real parts**

$$x(t) = \sum_{k=1}^N x_k(t) = \text{Re}\left\{\sum_{k=1}^N X_k e^{j\omega_0 t}\right\}$$

3 **Factor out the rotation**

$$x(t) = \text{Re}\left\{\left(\sum_{k=1}^N X_k\right) e^{j\omega_0 t}\right\}$$

4 **Return to cosine form**

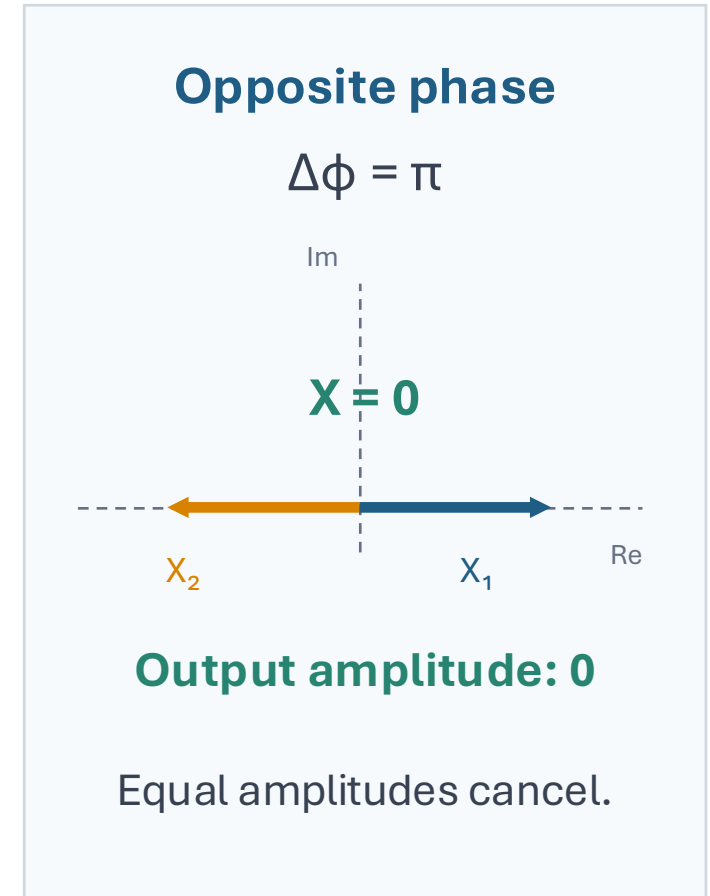
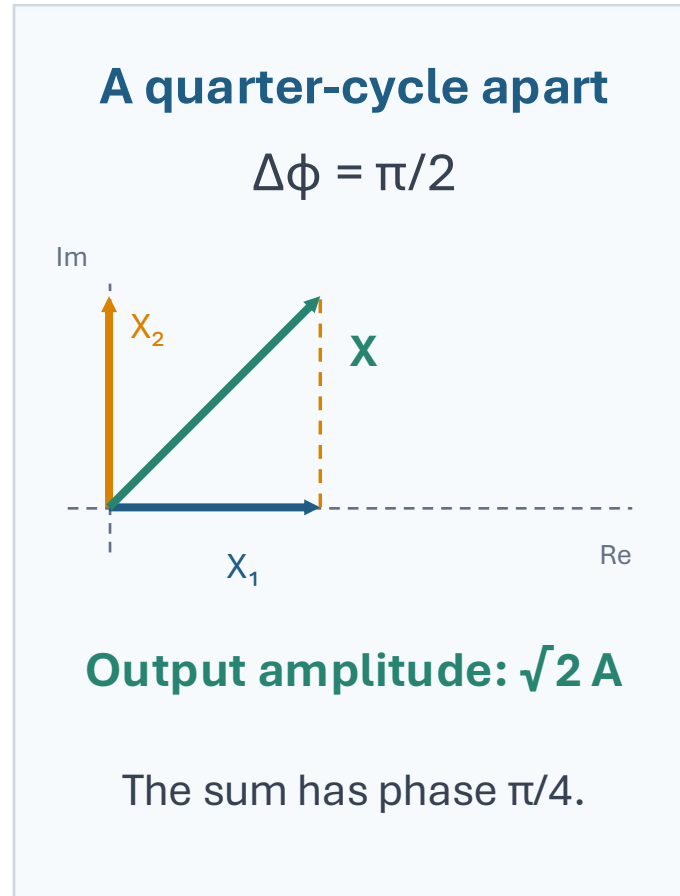
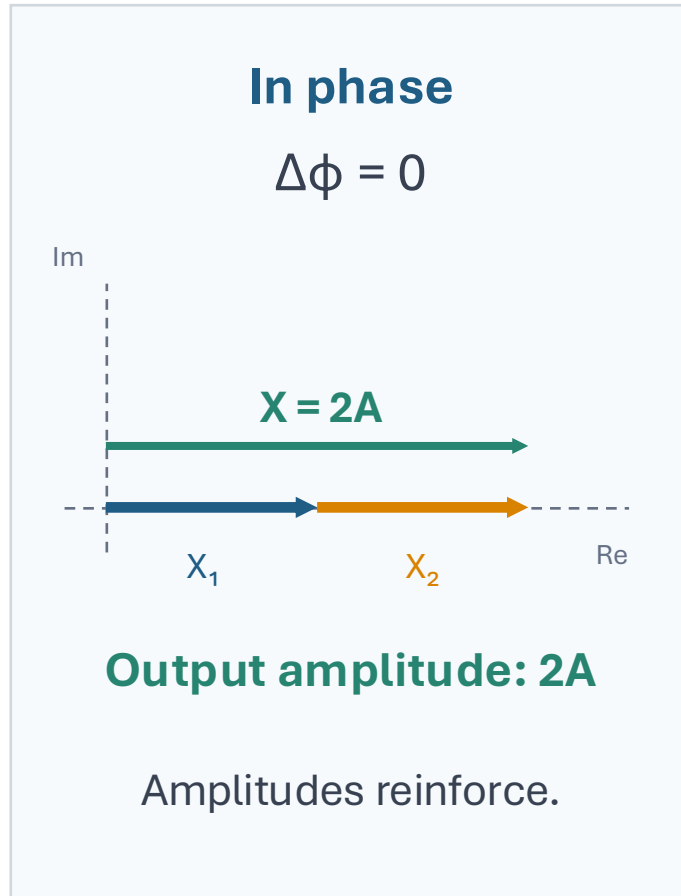
$$X = \sum_{k=1}^N X_k = A e^{j\varphi}$$

$$x(t) = \text{Re}\{A e^{j(\omega_0 t + \varphi)}\} = A \cos(\omega_0 t + \varphi)$$

Same frequency is essential: it lets us factor out one common rotation.

Why phase matters: reinforcement and cancellation

Two sinusoids at the same frequency, each with amplitude A



Add vectors, not amplitudes. Relative phase stays constant when frequencies match.

Example 3: adding two sinusoids

Use the phasor method instead of trigonometric expansion

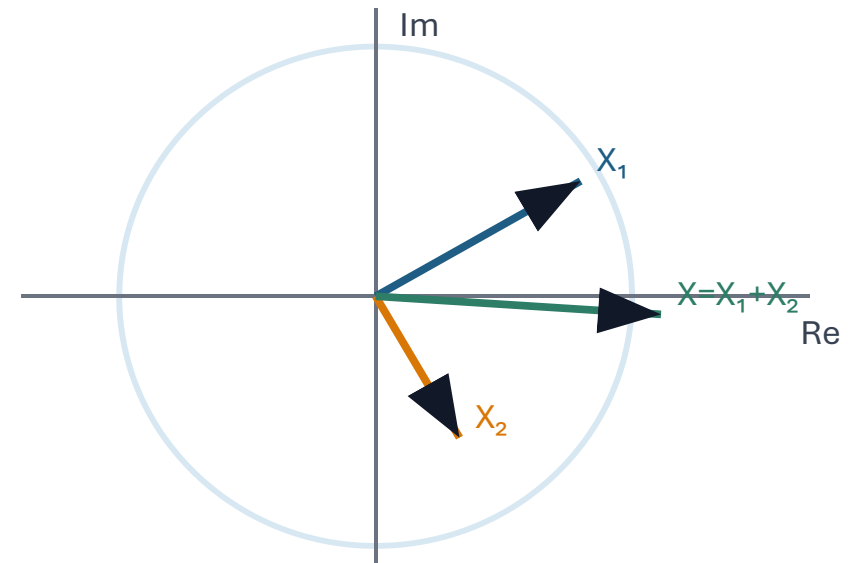
$$x(t) = 3\cos(\omega t + \pi/6) + 2\cos(\omega t - \pi/3)$$

Phasors

$$X_1 = 3e^{j\pi/6}, \quad X_2 = 2e^{-j\pi/3}$$

$$X = X_1 + X_2 \approx 3.598 - j0.232$$

$$|X| \approx 3.61, \quad \angle X \approx -0.065 \text{ rad}$$



$$x(t) \approx 3.61\cos(\omega t - 0.065)$$

Notation and takeaways

The minimum language for working with sinusoids

Symbol	Meaning
A	Amplitude / peak value
ω_0	Radian frequency, rad/s
f_0	Frequency in hertz, Hz
ϕ	Initial phase angle
T_0	Fundamental period, $T_0=1/f_0$
T_s	Sampling period
X	Complex amplitude or phasor

Three core takeaways

- A sinusoid is fully described by amplitude, frequency, and phase.
- Sampling evaluates the continuous waveform at $t = nT_s$.
- Complex phasors make sinusoidal algebra graphical and linear.

sinusoids + phasors = simpler signal analysis

Takeaway summary: from waves to phasors

Four ideas to remember — and use when solving problems

1 Describe the wave

Amplitude, frequency and phase define a sinusoid.

$$x(t) = A \cos(\omega_0 t + \varphi)$$

$$\omega_0 = 2\pi f_0, \quad T_0 = 1/f_0$$

2 Translate phase into delay

A positive time delay gives a negative phase shift.

$$\varphi = -\omega_0 t_1$$

3 Sample the waveform

Sampling evaluates the wave at discrete times.

$$x[n] = A \cos(\omega_0 n T_s + \varphi)$$

$T_s = 1/f_s$; phase is defined modulo 2π .

4 Add same-frequency sinusoids

Add complex phasors; then recover amplitude and phase.

$$X = \sum_{k=1}^N A_k e^{j\varphi_k}$$

$A = |X|$, $\phi = \arg X$ (when $X \neq 0$).

Key idea: complex exponentials turn sinusoidal analysis into vector algebra.